

SKILL – 22

Folder Setup and Tool Installation:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ valgrind --version
valgrind-3.22.0
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ |
```

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main(void) {
    char *message = malloc(50);

    if (message == NULL) {
        fprintf(stderr, "Memory allocation failed\n");
        return 1;
    }

    strcpy(message, "OS Skill 22: Memory Debugging");
    printf("%s\n", message);

    /* Intentional bug: memory is not freed. */
    return 0;
}
```

Run Valgrind and identify the memory leak:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ valgrind --leak-check=full --show-leak-kinds=all ./memory_bug
==3977== Memcheck, a memory error detector
==3977== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==3977== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==3977== Command: ./memory_bug
==3977==
OS Skill 22: Memory Debugging
==3977==
==3977== HEAP SUMMARY:
==3977==    in use at exit: 50 bytes in 1 blocks
==3977==   total heap usage: 2 allocs, 1 frees, 1,074 bytes allocated
==3977==
==3977== 50 bytes in 1 blocks are definitely lost in loss record 1 of 1
==3977==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==3977==    by 0x10919E: main (memory_bug.c:7)
==3977==
==3977== LEAK SUMMARY:
==3977==    definitely lost: 50 bytes in 1 blocks
==3977==    indirectly lost: 0 bytes in 0 blocks
==3977==    possibly lost: 0 bytes in 0 blocks
==3977==    still reachable: 0 bytes in 0 blocks
==3977==    suppressed: 0 bytes in 0 blocks
==3977==
==3977== For lists of detected and suppressed errors, rerun with: -s
==3977== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
```

Fix the memory leak and release resources:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ nano memory_fixed.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ gcc -Wall -Wextra -g memory_fixed.c -o memory_fixed
./memory_fixed
OS Skill 22: Memory Debugging
Memory released successfully
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ valgrind --leak-check=full --show-leak-kinds=all ./memory_fixed
==3989== Memcheck, a memory error detector
==3989== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==3989== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==3989== Command: ./memory_fixed
==3989==
OS Skill 22: Memory Debugging
Memory released successfully
==3989==
==3989== HEAP SUMMARY:
==3989==    in use at exit: 0 bytes in 0 blocks
==3989==   total heap usage: 2 allocs, 2 frees, 1,074 bytes allocated
==3989==
==3989== All heap blocks were freed -- no leaks are possible
==3989==
==3989== For lists of detected and suppressed errors, rerun with: -s
==3989== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ |
```

Design test cases and create an automated test script:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ ./test_script.sh
=====
OS SKILL 22 - AUTOMATED TESTS
=====
OS Skill 22: Memory Debugging
Memory released successfully
TEST 1: Program output - PASS
Running Valgrind memory test...
TEST 2: Memory errors - PASS
TEST 3: Memory cleanup - PASS
=====
ALL TESTS PASSED
Report saved to test_report.txt
Valgrind details saved to valgrind_report.txt
```

Compare results and generate the test report:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ cat test_report.txt
OS SKILL 22 - TEST REPORT
-----
Test 1: Program output - PASS
Test 2: Memory errors - PASS
Test 3: Memory cleanup - PASS
Overall Result: ALL TESTS PASSED
```

Detailed Valgrind Report After Automated Testing:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ cat valgrind_report.txt
==4005== Memcheck, a memory error detector
==4005== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==4005== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==4005== Command: ./memory_fixed
==4005==
OS Skill 22: Memory Debugging
Memory released successfully
==4005==
==4005== HEAP SUMMARY:
==4005==    in use at exit: 0 bytes in 0 blocks
==4005==    total heap usage: 2 allocs, 2 frees, 4,146 bytes allocated
==4005==
==4005== All heap blocks were freed -- no leaks are possible
==4005==
==4005== For lists of detected and suppressed errors, rerun with: -s
==4005== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ grep -E "HEAP SUMMARY|in use at exit|LEAK SUMMARY|ERROR S
UMMARY|All heap blocks" valgrind_report.txt
==4005== HEAP SUMMARY:
==4005==    in use at exit: 0 bytes in 0 blocks
==4005== All heap blocks were freed -- no leaks are possible
==4005== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$
```

Project Files and Documentation:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$ ls -l
total 68
-rw-r--r-- 1 bhardwaj bhardwaj 1074 Oct 10 06:39 README.md
-rw-r--r-- 1 bhardwaj bhardwaj 1 Oct 7 14:08 hi
-rwxr-xr-x 1 bhardwaj bhardwaj 18576 Oct 10 06:33 memory_bug
-rw-r--r-- 1 bhardwaj bhardwaj 366 Oct 10 06:34 memory_bug.c
-rwxr-xr-x 1 bhardwaj bhardwaj 18672 Oct 10 06:37 memory_fixed
-rw-r--r-- 1 bhardwaj bhardwaj 405 Oct 10 06:35 memory_fixed.c
-rw-r--r-- 1 bhardwaj bhardwaj 174 Oct 10 06:37 test_report.txt
-rwxr-xr-x 1 bhardwaj bhardwaj 1446 Oct 10 06:36 test_script.sh
-rw-r--r-- 1 bhardwaj bhardwaj 668 Oct 10 06:37 valgrind_report.txt
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-22$
```